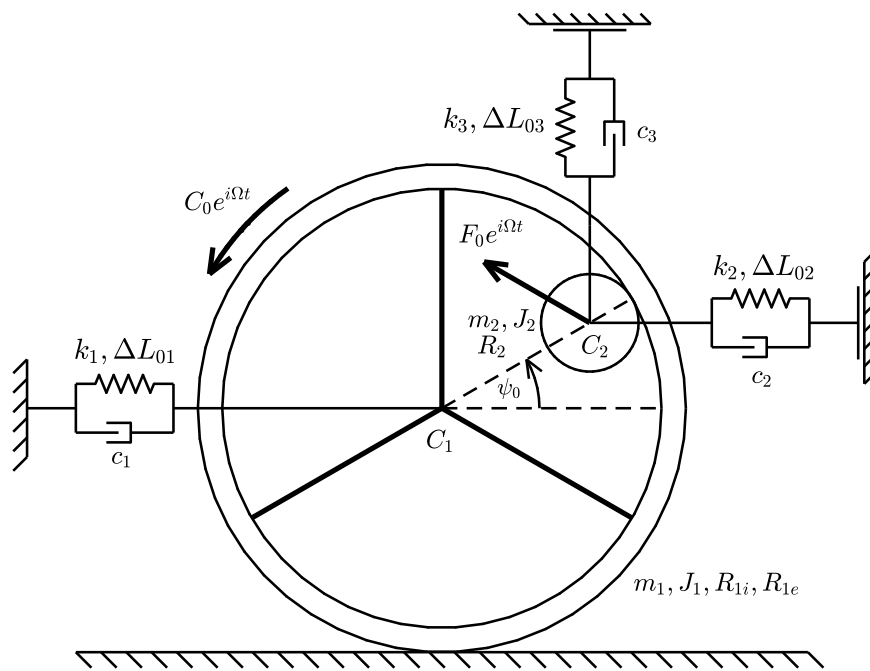


Dynamics of Mechanical Systems
January 09th, 2019
Standard problem (grading up to 30/30)



$m_1 = 15 \text{ kg}$	$k_1 = 20\text{e}3 \text{ N/m}$
$m_2 = 5 \text{ kg}$	$k_2 = 10\text{e}3 \text{ N/m}$
$J_1 = 0.6 \text{ kgm}^2$	$k_3 = 20\text{e}3 \text{ N/m}$
$J_2 = 0.05 \text{ kgm}^2$	$c_1 = 6 \text{ Ns/m}$
$R_{1e} = 0.5 \text{ m}$	$c_2 = 2 \text{ Ns/m}$
$R_{1i} = 0.4 \text{ m}$	$c_3 = 2 \text{ Ns/m}$
$R_2 = 0.1 \text{ m}$	$\psi_0 = \pi/6$
$\Delta L_{01} = 0.0824 \text{ m}$	
$\Delta L_{02} = 0.1647 \text{ m}$	
$\Delta L_{03} = 0.05 \text{ m}$	

The mechanical system above lays in the vertical plane and is shown in its equilibrium position. It is composed by a disk with mass m_1 , external and internal radii R_{1e} and R_{1i} respectively, that rolls without sliding on a horizontal plane. A second disk with mass m_2 and radius R_2 rolls without sliding on the internal radius R_{1i} of the first disk. A spring and damper system (k_1, c_1) is connected to the disk center C_1 of the first disk, and two spring and damper systems (k_2, c_2) (k_3, c_3) are connected to the disk center C_2 of the second disk. In the equilibrium position the angle between the horizontal direction and the line C_1C_2 is ψ_0 . A harmonic torque is applied to the first disk, and a harmonic force is applied in the center of the second disk; the force is inclined by $2/3 \pi$ with respect to the horizontal direction, positive counterclockwise.

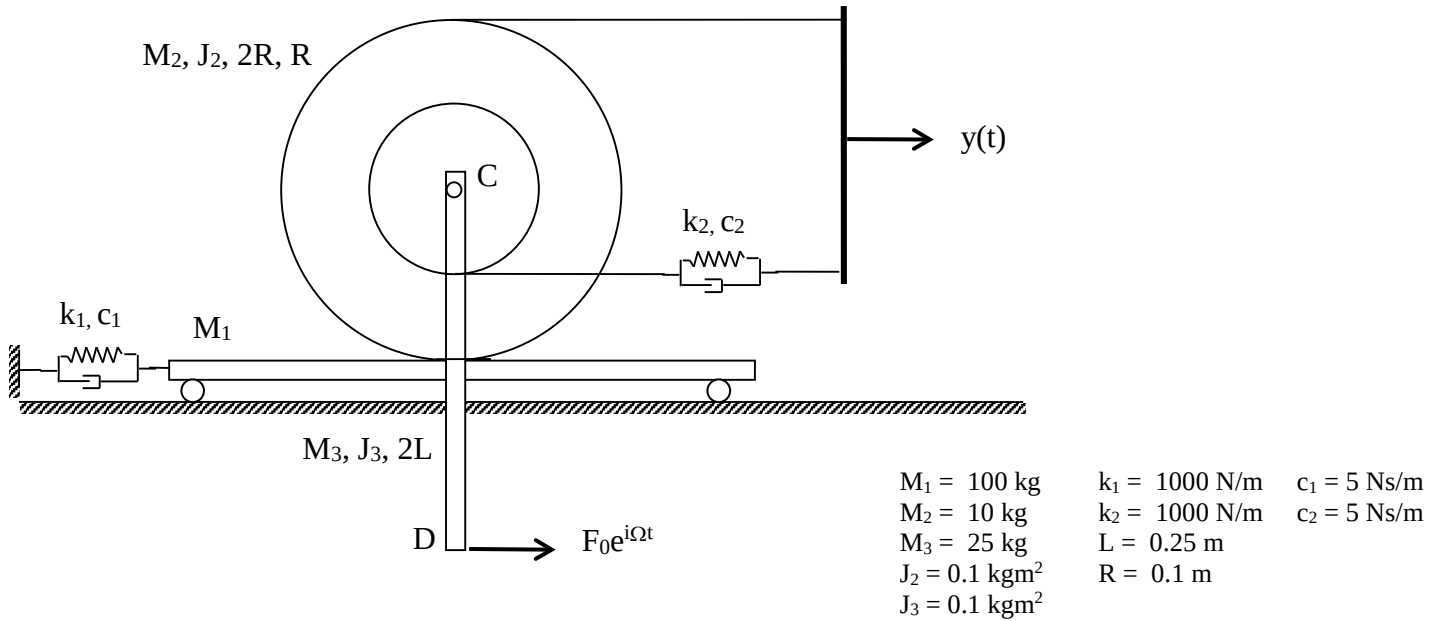
The students are requested to:

1. Write their surname, name, identification number and SNxxx code (see note below) on the top of first page of each sheet, leaving blank field "Nr." Sheets with uncompleted personal data will not be considered. For environmental reasons the students are kindly requested to use all of the four pages of each sheet before using another one;
2. Write the linearized equations of motion around the given equilibrium configuration;
3. Compute the natural frequencies and modes of vibrations. This answer shall be provided in written on paper;
4. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic torque $C_0 e^{i\Omega t}$. Output: the rotation of the disk m_1 and the relative rotation of the disk m_2 with respect to the rotation of the disk m_1 . Save the graphical results as SNxxx1.fig and SNxxx2.fig, with S the first letter of your surname, N the first letter of your name and xxx are the last three digits of your matriculation number (e.g. for Bruni Stefano 123456: BS4561.fig and BS4562.fig)
5. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic force $F_0 e^{i\Omega t}$. Output: the vertical component of the displacement of point C_2 and the elastic and damping force transmitted by the spring and damper system (k_2, c_2). Save the graphical results as SNxxx3.fig and SNxxx4.fig
6. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic torque $C_0 e^{i\Omega t}$. Output: the horizontal and vertical dynamic component of the contact force between the disk m_1 and the horizontal plane. Save the graphical result as SNxxx5.fig and SNxxx6.fig;
7. Find the minimum amplitude of the harmonic torque $C_0 e^{i\Omega t}$ that leads to a loss of contact of disk 1 from the horizontal constraint, for $\Omega = 2\pi f$, $f = 2\text{Hz}$. This answer shall be provided in written on paper

Note: Please collect the Matlab's script file relative to the numerical solution of the exercise and save it with name SNxxx.m, with S the first letter of the surname, N the first letter of the name and xxx the last three digits of the matriculation number (e.g. for Bruni Stefano 123456: BS456.m). It is mandatory that all of the files (fig e .m) are collected in one single compressed folder (in .zip or.rar format) named SNxxx.zip (or SNxxx.rar), which has to be uploaded on Beep. To upload the sipped folder:

1. Connect to the site <https://beep.metid.polimi.it> and enter the portal using your credentials;
3. Select section **Homework** of the course DYNAMICS OF MECHANICAL SYSTEMS (BRUNI STEFANO);
4. Select the folder *Test January 09th 2019 L06* or *Test January 09th 2019 L07* according to your classroom
5. Click on button Add and select Single document;
6. Browse the compressed file SNxxx.zip using the button Browse... (or Sfogliala...)
7. Fill the "Title" field with your code SNxxx (e.g. for Bruni Stefano 123456: BS456);
8. Confirm the procedure clicking the button Publish.

Dynamics of Mechanical Systems
January 09th, 2019
Simplified problem (grading up to 22/30)



The mechanical system above, shown in its equilibrium position, lays in the vertical plane and it is constituted by:

- A double disk M_2 (the smaller disk is fixed to the bigger one in the same center), rolling without sliding on the cart M_1 ;
- A rigid bar CD connected with a hinge to the disk center C .

The students are requested to:

1. Write the surname, name, identification number and SNxxx code (see note below) on the top of first page of each sheet and leave blank the field named *Nr*. The sheet with uncompleted personal data may be not corrected. For environmental reasons the students are also kindly requested to write all of the four pages of each sheet before using another one.
2. Write the linearized equations of motion around the given static configuration;
3. Compute the natural frequencies and modes of vibrations;
4. Compute the Frequency Response Function (in the range 0 – 2 Hz, step = 0.01 Hz) applying the horizontal force $F_0 e^{i\Omega t}$. Output: the cart displacement and the double disk rotation. Save the graphical results as SNxxx1.fig and SNxxx2.fig, with S the first letter of your surname, N the first letter of your name and xxx are the last three digits of your identification number (e.g. for Bruni Stefano 123456: BS4561.fig and BS4562.fig)
5. Compute the Frequency Response Function (in the range 0 – 2 Hz, step = 0.01 Hz) applying an harmonic displacement $y(t) = y_0 e^{i\Omega t}$. Output: the horizontal displacement of the center of mass of the CD bar and the force transmitted by the spring and damper system (k_2 c_2). Save the graphical results as SNxxx3.fig and SNxxx4.fig

Note: Please collect the Matlab's script file relative to the numerical solution of the exercise and save it with name SNxxx.m, with S the first letter of the surname, N the first letter of the name and xxx the last three digits of the matriculation number (e.g. for Bruni Stefano 123456: BS456.m). It is mandatory that all of the files (fig e .m) are collected in one single compressed folder (in .zip or.rar format) named SNxxx.zip (or SNxxx.rar), which has to be uploaded on Beep. To upload the sipped folder:

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